

Friday 28th August, 2026

Dr. Mauricio González-Forero
Senior Postdoctoral Fellow — Theoretical Evolutionary Biologist
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Citizenship: British and Colombian.

Professional Profile and Research Vision

I am a theoretical evolutionary biologist specializing in the integration of evolutionary and developmental (evo-devo) dynamics and advanced simulation-based statistical inference. I am interested in formulating mathematical theory combining evolutionary-developmental dynamical systems, empirical data, and Bayesian inference to both infer evolutionary causes and predict the effects of developmental and evolutionary intervention. A particular focus is the question of why the human brain size evolved.

Core Methodological Expertise

- **Evolutionary Methods:** Advanced proficiency in the mathematical theories of Population Genetics, Quantitative Genetics, Adaptive Dynamics, Social Evolution, Evolutionary Game Theory.
- **Mathematical Toolkit:** Deterministic and Stochastic Dynamical Systems, Optimal Control, Mathematical Statistics, Ordinary and Partial Differential Equations, Matrix Algebra, Matrix Calculus, Set Theory.
- **Statistical Inference:** Simulation-Based Inference, Approximate Bayesian Computation, Parameter Estimation, Sensitivity Analysis, Bayesian Model Selection.
- **Programming Languages:** Advanced proficiency in computational simulation and mathematical scripting (Julia, Mathematica, MATLAB, R, $\LaTeX$).

Academic Appointments

2025-2028	Senior Postdoctoral Fellow Konrad Lorenz Institute for Evolution and Cognition Research, Klosterneuburg, Austria. Focus: Advancing a mathematical framework integrating development and evolution.
2018-2024	Research Fellow Supervisor: Andy Gardner. University of St Andrews, UK. Focus: Mathematical modelling of the evolutionary and developmental (evo-devo) dynamics of the human brain size.
2020-2021	$\approx$ 3.8 months of furlough due to increased childcare responsibilities during the COVID pandemic. Overall, I estimate the pandemic delayed my work by $\approx$ 7 months.

2016-2018	Marie Skłodowska-Curie Individual Fellow Supervisor: Andy Gardner. University of St Andrews, UK. Focus: Mathematical modelling human brain size evolution under social interactions.
2013-2016	Postdoctoral researcher Supervisor: Laurent Lehmann. University of Lausanne, Switzerland. Focus: Mathematical modelling human brain size evolution without social interactions.
2010-2012	Graduate research associate Funded by the National Institute of Mathematical and Biological Synthesis (NIM-BioS). University of Tennessee, Knoxville, USA. Focus: Mathematical modelling the evolution of eusociality via maternal manipulation

Education

2007-2013	PhD in Ecology and Evolutionary Biology University of Tennessee, Knoxville, USA. Thesis: Evolution of acquiescence to manipulation. Advisor: Sergey Gavrilets. Specialised postgraduate coursework (Mathematics and Statistics): Mathematical Ecology I-II, Mathematical Evolutionary Theory, Stochastic Modelling (Markov Chains), and Statistical Inference I-II.
1999-2006	BSc in Biology (and the equivalent of a minor in Mathematics) University of Antioquia, Medellín, Colombia. Thesis: Species: a mathematical formalization. Specialised undergraduate coursework (Mathematics and Physics): Calculus I-III, Linear Algebra, Multilinear Algebra, Ordinary Differential Equations, Vectorial Geometry, Mathematical Logic, Set Theory, and Physics I.

Key Research Contributions

- *In-Silico* Brain Evolution: Developed novel mathematical methods that replicate human brain and body size evolution *in silico* integrated with empirical data, offering the opportunity to launch a new research avenue of Bayesian simulation-based inference of the drivers of brain evolution in a species of choice.
- Mathematical Synthesis of Development and Evolution: Formulated a mathematical integration of developmental and evolutionary dynamics with realistic sexual genetics, opening the door to a new era of advanced evolutionary understanding.
- Social Evolution: Formulated rigorous mathematical models showing how ancestral manipulative behaviours can lead to voluntary altruism, offering a shift in paradigm in behavioural ecology.
- Translational Quantitative Applications: Applied my mathematical developments to predict demographic interventions that can extend human healthy lifespan.

Teaching experience

2018-2019, 2023	Guest lecturer Designed and gave 1 lecture on Brain evolution: comparative neuroanatomy and controversies (3rd year Evolution module). Gave 3 lectures on Social Evolution (3rd year Evolution module). Also essay setting, marking, and providing feedback (2nd and 3rd Evolution modules). University of St Andrews, UK.
2013-2016	Teaching assistant Grading master-student projects, mentoring students in the Experimental Design module, and assisting exercise sections of the Population Genetics module. University of Lausanne, Switzerland.
2010	In charge of a full module Designed and gave all lectures, exams, and grading of the semester-long module. Module: Mathematics for Life Sciences. Approx. 30 students from majors including Medicine and Psychology. University of Tennessee, Knoxville, USA.
2007-2009	Graduate teaching assistant Teaching wet lab sections of the modules Biology, Genetics, and Mathematics for Life Sciences. University of Tennessee, Knoxville, USA.

Preprints

González-Forero M. A mathematical synthesis of genetics, development, and evolution. In revision at *Genetics*. Preprint: <https://www.biorxiv.org/content/10.64898/2026.02.25.707927v2>.

González-Forero M, Gómez-Robles A. Mechanistic quantification of the effects of genes, environment, and development on human behaviour. In revision for *Evolutionary Human Sciences*. Preprint: <https://www.biorxiv.org/content/10.1101/2025.03.17.643651v4>.

Riffe T, Villavicencio F, **González-Forero M**. Sensitivity and decomposition of multistate healthy life expectancy. In revision at *Demography*. Preprint: <https://dx.doi.org/10.4054/MPIDR-WP-2024-014>.

Peer-reviewed publications

*Three selected publications, with key contribution noted.

González-Forero M, Gómez-Robles A. (2025) Why did the human brain size evolve? A way forward. *Phil. Trans. B.* 380, 20240114. Open access. <https://royalsocietypublishing.org/doi/10.1098/rstb.2024.0114>.

***González-Forero M.** (2024) Evolutionary-developmental (evo-devo) dynamics of hominin brain size. *Nature Human Behaviour.* 8, 1321–1333. Open access. <https://www.nature.com/articles/s41562-024-01887-8>.

Key contribution: Established a method to analytically model the evolutionary and developmental dynamics of human brain size.

***González-Forero M.** (2024) A mathematical framework for evo-devo dynamics. *Theoretical Population Biology.* 155, 24-50. Open access. <https://www.sciencedirect.com/science/article/pii/S0040580923000758>.

Key contribution: Formulated a mathematical method to synthesise and analyse evolutionary and developmental dynamics.

- ***González-Forero M.** (2023) How development affects evolution. *Evolution*. 77, 562-579. Open access. <https://academic.oup.com/evolut/article/77/2/562/6955321>.
Key contribution: Established fundamental implications of my mathematical synthesis of development and evolution.
- González-Forero M**, Peña J. (2021) Eusociality through conflict dissolution. *Proc. R. Soc. B*. 288, 20210386. Open access. <https://royalsocietypublishing.org/doi/10.1098/rspb.2021.0386>.
- González-Forero M**, Gardner A. (2018) Inference of ecological and social drivers of human brain-size evolution. *Nature* 557, 554-557. <https://www.nature.com/articles/s41586-018-0127-x>.
Accessible PDF: <https://rdcu.be/O1Vc>.
- González-Forero M**, Faulwasser T, Lehmann L. (2017) A model for brain life history evolution. *PLoS Computational Biology* 13, e1005380. Open access. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005380>.
- González-Forero M.** (2015) Stable eusociality via maternal manipulation when resistance is costless. *Journal of Evolutionary Biology* 28, 2208-2223. <https://academic.oup.com/jeb/article/28/12/2208/7412264>. Accessible PDF: <https://mauriciogforero.github.io/FullJEB15.pdf>.
- González-Forero M.** (2014) An evolutionary resolution of manipulation conflict. *Evolution* 68, 2038-2051. <https://academic.oup.com/evolut/article/68/7/2038/6852554>. Accessible PDF: <https://mauriciogforero.github.io/FullEvolution14.pdf>.
- González-Forero M**, Gavrillets S. (2013) Evolution of manipulated behavior. *American Naturalist* 182, 439-451. <https://www.journals.uchicago.edu/doi/10.1086/671932>. Accessible PDF: <https://mauriciogforero.github.io/AmNat13.pdf>.
- González-Forero M.** (2009) Removing ambiguity from the biological species concept. *Journal of Theoretical Biology* 256, 76-80. <https://www.sciencedirect.com/science/article/abs/pii/S002251930800489X>. Accessible PDF: <https://mauriciogforero.github.io/FullJTB09.pdf>.

Non-peer-reviewed publications

- González-Forero M.** (2026) From nerve cells to consciousness. In Roberts A. (Ed.) *Humans: The Evolution of a Species*. DK. <https://geni.us/Humans-DK>.
- González-Forero M.** (2026) Growing brains. In Roberts A. (Ed.) *Humans: The Evolution of a Species*. DK. <https://geni.us/Humans-DK>.
- Pim Edelaar, Niels J Dingemanse, Samantha C. Patrick, Denis Réale, Alastair J. Wilson, Amélie Baud, Daniel Bolnick, Dries Bonte, Raul Costa Pereira, Troy Day, Laurel Fogarty, Willem Frankenhuis, **Mauricio González-Forero**, Marie I. Kaiser, Julien G. A. Martin, Jason Matthiopoulos, Klaus Reinhold, Holger Schielzeth, Meike J. Wittmann, Jonathan Wright, Melanie Dammhahn (2026) Towards a better understanding of adaptation: Problem description, partial solutions, and recommendations. <https://doi.org/10.32942/X2108Z>.
- González-Forero M.** (2018) The human brain from ecology and seemingly culture. Behind the paper blog post at *Nature Ecology and Evolution Community*. <https://communities.springernature.com/posts/the-human-brain-from-ecology-and-seemingly-culture>.
- González-Forero M.** (2018) Why do humans have such large brains? Our study suggests ecology was the driving force. *The Conversation*. <https://theconversation.com/why-do-humans-have-such-large-brains-our-study-suggests-ecology-was-the-driving-force-96873>.

Research Grants, Awards, and Honours

	Total awarded (at current exchange rates): $\approx$ € 219k.
2016-2018	Marie Skłodowska-Curie Individual Fellowship (€ 183,455) Project title: Brain growth under social pressure: mathematical modelling of brain growth when individuals face social challenges. Awarded by the EU European Commission.
2010-2012	Graduate Research Assistantship (US\$ 40,000) Project title: Evolution of eusociality mediated by maternal induction. Awarded by the USA National Institute for Mathematical and Biological Synthesis, NIMBioS.
2009	Santa Fe Institute Complex Systems Summer School (US\$ 550) Awarded by the Santa Fe Institute, Santa Fe, NM, USA
2006	Undergraduate thesis award Thesis title: "Species: a mathematical formalization". Awarded by the Faculty of Natural and Exact Sciences, University of Antioquia, Colombia.
2005	Third place in the National Biology Exam (ECAES) Exam for all undergraduates in Colombia. Awarded by the Ministry of Education, Colombia.

Consultancy

2023-2025	Private consultancy (€ 20,000) Sensitivity of human life expectancy to perturbation. University of Basque Country, Spain.
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Invitations

2026	Invited talk and workshop participation Invited to the 45th Altenberg Workshop in Theoretical Biology titled "The Developmental Origins of Evolvability" https://www.kli.ac.at/en/events/kli_workshops/view/744 Konrad Lorenz Institute for Evolution and Cognition Research, Austria.
	Invited talk Department of Biology, Lund University, Sweden.
	Invited talk Department of Zoology, Stockholm University, Sweden.
2025	Invited talk Modeller's Seminar, The Francis Crick Institute, London, UK.
	Invited talk Department of Anthropology, University College London, UK.
	Invited talk Nick Barton lab meeting, Institute of Science and Technology (ISTA), Austria.
	Invitation to workshop Invited to the Volkswagen Foundation Scoping Workshop "Building a framework to integrate diverse routes to adaptation in a changing world" organised by Melanie Dammhahn, Niels Dingemanse, Pim Edelaar, Sam Patrick, Denis Réale, and Alastair Wilson. Hannover, Germany.
	Invited talk

	Willem Frankenhuis lab meeting. University of Amsterdam. Online.
2024	Invited talk School of Psychology & Neuroscience, University of St Andrews, UK.
	Invited talk Symposium Connecting Macro and Micro Evolution of Form and Function Relationships at the Evolution Conference, Montreal, Canada (given virtually).
	Invited talk Royal Society meeting “Natural Selection Shapes Animal Cognition” in Bath, UK, organised by Elli Leadbeater and Alex Thornton.
	Invited paper Special issue of <i>Philosophical Transactions of the Royal Society B</i> “Natural Selection Shapes Animal Cognition”.
2023	Invited talk Institute Seminar. Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany.
	Invited talk Roberto Salguero-Gómez lab meeting, Department of Biology, University of Oxford, UK. Online.
	Invited talk Applied Mathematics Seminar series, School of Mathematics & Statistics, University of St Andrews, UK.
2022	Invited talk Research Away Day, School of Biology, University of St Andrews, UK.
2020	Invited talk Institute for Advanced Studies, Toulouse, France. Online due to pandemic.
2019	Invited talk Evolutionary Anthropology Research Group, Department of Anthropology, Durham University, UK.
	Invited talk Applied Mathematics Seminar series, School of Mathematics & Statistics, University of St Andrews, UK.
2018	Invited talk Stochastic Models for the Inference of Life Evolution group at the Collège de France in Paris.
2017	Invited talk Department of Zoology, University of Cambridge.
	Invited talk Department of Human Behavior, Ecology and Culture at the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany.
2016	Invited talk Centre for Biological Diversity, School of Biology, the University of St Andrews, UK.
2015	Invited talk Department of Evolutionary Theory, Max Planck Institute for Evolutionary Biology, Plön, Germany.

Organised workshops and symposia

2026	Co-organiser of the symposium “Advances in theoretical evo-devo”. Conference of the European Society for Evolutionary Developmental Biology (Euro Evo Devo), Glasgow, UK.
2020	Co-organiser of Fellowship writing workshop for the School of Biology. 10 speakers from the University of St Andrews. University of St Andrews, UK.

2019	Co-organiser of the symposium “Life history evolution: bridging theory and data”. Congress of the European Society for Evolutionary Biology (ESEB2019), Turku, Finland.
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Administrative experience

2026	Student poster evaluator. Conference of the European Society for Evolutionary Developmental Biology (Euro Evo Devo), Glasgow, UK.
2022-2024	Postdoctoral representative of the School of Biology for the Institute of Behavioural and Neural Sciences (IBANS), University of St Andrews.
2019-2021	Postdoctoral representative for the Biology Equality and Diversity Committee, School of Biology, University of St Andrews.

Reviewer for

American Naturalist, Biological Reviews, Biological Theory, Bulletin of Mathematical Biology, eLife, Evolution, Evolution Letters, Genetics, iScience, Journal of Evolutionary Biology, Journal of Theoretical Biology, Journal of Zoology, Nature, Nature Communications, Nature Ecology and Evolution, PLoS ONE, Proceedings of the Royal Society B, Science Advances, Theoretical Population Biology.

Grants reviewed for

Research Foundation - Flanders (Belgium).

Contributed conference presentations

2026	Talk and poster. Conference of the European Society for Evolutionary Developmental Biology (Euro Evo Devo), Glasgow, UK.
	Talk. Mathematical Models in Ecology and Evolution conference. Cork, Ireland.
2025	Talk and poster. European Society for Evolutionary Biology conference. Barcelona, Spain.
	Talk. EvolVienna meeting. Vienna, Austria.
2024	Virtual talk. Conference of Australasian Society for Human Biology.
	Virtual talk. Evolution conference.
2022	Talk. Mathematical Models in Ecology and Evolution conference. Reading, UK.
	Talk. Euro Evo-Devo. Naples, Italy.
	Talk. Population Genetics Group. Online.
2021	Talk. Evolution conference. Online.
2019	Talk. Mathematical Models in Ecology and Evolution conference. Lyon, France.
	Talk. European Human Behaviour and Evolution Association conference. Toulouse, France.
2018	Talk. Models in Population Dynamics, Ecology, and Evolution conference. Leicester, UK.
2017	Talk. European Human Behaviour and Evolution Association conference. Paris, France.
2016	Talk. Institute of Behavioural and Neural Sciences mini symposium. St Andrews, UK.
	Talk. European section of International Union for the Study of Social Insects conference. Helsinki, Finland.
	Talk. Swiss Biology '16 conference. Lausanne, Switzerland.

2015	2 posters. European Society for Evolutionary Biology conference. Lausanne, Switzerland.
	Talk. Mathematical Models in Ecology and Evolution conference. Paris, France.
	Poster. 3rd Economics and Biology Workshop of the Institute of Advanced Study. Toulouse, France.
2012	Talk. Society for Mathematical Biology conference. Knoxville, TN, USA.
	Talk. Evolution conference. Ottawa, Canada.
2011	Talk. Evolution conference. Norman, OK, USA.
2010	Talk. BiocompletiX ^{XI} conference: The evolution of cooperation. Bloomington, IN, USA.
2007	Talk. South Eastern Population Ecology and Evolutionary Genetics meeting. Tremont, TN, USA.
2005	Poster. European Conference on Mathematical and Theoretical Biology. Dresden, Germany.

Attended workshops and Summer schools

2017	2 workshops. Mean field games, and Numerical methods for optimal control problems. Sapienza University. Rome, Italy.
2015	Workshop. 3rd Economics and Biology Workshop, Institute for Advanced Study. Toulouse, France.
2014	Workshop. Social evolution: merits and limitations of inclusive fitness theory. Arolla, Switzerland.
2010-2012	Working group. Large scale demographic, network and behavioral trait analyses of sociality. Four meetings at National Evolutionary Synthesis Center, Durham, USA.
2009	Summer Program. Mathematical Biosciences Institute Summer Graduate Program. Columbus, OH, USA.
	Complex Systems Summer School. Santa Fe Institute. Santa Fe, NM, USA.
2004	Workshop. Bases in Studies of Entomologic Diversity in Ibero-America. Quito and Yasuni National Park (in the Amazon rainforest), Ecuador.

Additional funding

2015	Travel grant. €500 by the Institute for Advanced Study, Toulouse, France to attend the Institute's 3rd Economics and Biology Workshop.
2012	Travel grants. US\$1000 by the National Institute of Mathematical and Biological Synthesis and US\$350 by the Department of Ecology and Evolutionary Biology, University of Tennessee, Knoxville to attend the Evolution conference in Ottawa, Canada.
2011	Travel grants. US\$500 by the National Institute of Mathematical and Biological Synthesis, US\$200 by the Graduate Researchers in Ecology, Behavior and Evolution of UTK, and US\$200 by the Department of Ecology and Evolutionary Biology to attend the Evolution conference in Norman, OK, USA.
2010	Travel grants. US\$400 by the workshop committee and US\$500 by the National Institute of Mathematical and Biological Synthesis to attend BiocompletiX ^{XI} .
	Travel grant. US\$650 by NIMBioS to visit Gro Amdam and James Hunt at Arizona State University.
2009	Travel grant and stipend. US\$400 of travel support and a stipend of US\$300 by the Mathematical Biosciences Institute to attend their Summer Graduate Program.

	Travel grant. US\$800 by the Department of Ecology and Evolutionary Biology of the University of Tennessee, Knoxville, to attend the Santa Fe Institute Summer School and the Mathematical Biosciences Institute Summer Program. Declined.
2005	Undergraduate thesis improvement grant. $\approx$ US\$30 by the Research Development Committee (CODI) of the University of Antioquia.
2004	Travel grant. US\$1040 by the Society for Mathematical Biology and €950 by the European Society of Mathematical and Theoretical Biology to attend the European Conference on Mathematical and Theoretical Biology and the Society of Mathematical Biology conference in Dresden, Germany.

Selected media coverage

2024	Helsingin Sanomat. https://www.hs.fi/tiede/art-2000010573820.html . PNAS journal club. https://www.pnas.org/post/journal-club/model-humans-got-their-big-brains-stirs-interest-and-controversy .
2023	I F*cking Love Science, among others. https://www.iflscience.com/we-may-finally-know-why-humans-have-such-big-brains-70453 .
2021	Brain evolution work featured in article by Science Illustrated (Sweden).
2019	Article by the European Commission research press. https://ec.europa.eu/research-and-innovation/en/projects/success-stories/all/why-humans-have-very-big-brains . Article by Discover Magazine. State of Science: Finding Human Ancestors in New Places. https://www.discovermagazine.com/planet-earth/state-of-science-finding-human-ancestors-in-new-places .
2018	Interviewed by The Associated Press (USA), the AFP (France), New Scientist (UK), Dagens Nyheter (Sweden), LA Times (USA), Vox (USA), La Vanguardia (Spain), TBS eFM Radio (Korea), Weekendavisen (Denmark), among others. Article by the Associated Press. Study offers new look at why our brains evolved to be so big. https://apnews.com/7fe4e81cbb114ee2adf42c6e49f19b45/Study-offers-new-look-at-why-our-brains-evolved-to-be-so-big . Article by LA Times. Can simulating evolution on a computer explain our enormous brains? http://www.latimes.com/science/sciencenow/la-sci-sn-human-brain-evolution-20180523-story.html . Article by Vox. Why do humans have such huge brains? https://www.vox.com/science-and-health/2018/5/23/17377200/human-brain-size-evolution-nature . Article by New Scientist. We may have got the evolution of our big brains entirely wrong. https://www.newscientist.com/article/2169862-we-may-have-got-the-evolution-of-our-big-brains-entirely-wrong/ . Radio interview by TBS eFM Radio This Morning program (Korea). http://cdn.podbbang.com/data1/tbsadm/thism180529002.mp3 .
2017	Interviewed by PBS Newshour (USA). Why did humans evolve big brains? We don't know, but math can help. http://www.pbs.org/newshour/rundown/humans-evolve-big-brains-dont-know-math-can-help/ . ScienceDaily. How big brains evolved could be revealed by new mathematical model. https://www.sciencedaily.com/releases/2017/03/170309142339.htm . EOS Wetenschap. Wiskundig model helpt de evolutie van grote breinen beter begrijpen. https://www.eoswetenschap.eu/psyche-brein/wiskundig-model-helpt-de-evolutie-van-grote-breinen-beter-begrijpen .

	ANSA Scienza&Tecnica. Un modello matematico ricostruisce l'evoluzione dei grandi cervelli. http://www.ansa.it/canale_scienza_tecnica/notizie/fisica_matematica/2017/03/10/un-modello-matematico-ricostruisce-levoluzione-dei-grandi-cervelli-_1be94a2f-bcb9-43d5-a0ea-e70c8e7d6e46.html .
2013	Nature World News. How manipulation becomes altruism: a study. http://www.natureworldnews.com/articles/3548/20130820/manipulation-becomes-altruism-study.htm .
	Business Standard. Altruism may have origins in manipulation. http://www.business-standard.com/article/news-ani/altruism-may-have-origins-in-manipulation-113082000973_1.html .
	Science World Report. The evolution of manipulation: altruism created by manipulated behaviour. http://www.scienceworldreport.com/articles/8899/20130819/evolution-manipulation-altruism-created-manipulated-behavior.htm .
	Der Standard. Manipulation könnte Wurzel für selbstloses Verhalten sein. http://derstandard.at/1376533943228/Manipulation-koennte-Wurzel-fuer-selbstloses-Verhalten-sein .